Hanyang Liu

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Education

University of Michigan, Ann Arbor — Ann Arbor, MI
M.S. Data Science — *Aug. 2025 – May 2027*

Stony Brook University — Stony Brook, NY
B.S. Applied Mathematics and Statistics (GPA: 4.0) — *Sep. 2021 – May 2025*

Professional Experience

Machine Learning Engineer Intern — Shenzhen, China
Shenzhen Linghang Technology Co., Ltd. — *Apr. 2026 – Aug. 2026*

- Contributed to an LLM customer-service assistant, integrating Tencent Hunyuan into a client's WeChat Mini Program.
- Built Python data-ingestion and cleaning scripts that parsed and normalized domain-specific data formats into structured inputs for the assistant, and tested the pipeline to catch format and edge-case errors before integration.

Research Experience

Strategic Reasoning Group, University of Michigan — Ann Arbor, MI
Graduate Researcher; Research Assistant — *Apr. 2026 – Present*

- Extended PyMarketSim, an agent-based limit-order-book simulator, with a wash-trading module built **entirely by subclassing** (no changes to the upstream matching engine). Paired orders at the spread midpoint reach a **99% self-trade rate** with near-zero price impact; **progressively evasive adversaries** (wallet rotation, randomized size, residual positions, pipelined execution) then stress-test each detector.
- Generated **150k+ ground-truth-labeled market simulations**, including a **200-seed, 104k-run sweep on the UMich Great Lakes HPC cluster (Slurm)**, with 100+ regression checks per version and cross-OS reproducibility checks.
- Showed that a published network detector **flags 40%+ of honest volume in active markets**; built a multi-channel detector (round-trip and repeat-trade evidence, 1% control-FPR calibration) raising active-market recall **0.17 → 0.65**, then a Louvain community channel that lifted recall against an adaptive adversary **0.30 → 0.82 at 0.93 precision** (within 0.03 on 100 held-out seeds).
  First-author paper in preparation: Studying Financial Market Pathologies via Agent-Based Simulation

Time Series Analytics Lab, Anhui University — Hefei, China
Undergraduate Researcher — *Nov. 2023 – Mar. 2025*

- Co-authored a VMD–Seq2Seq power-load forecasting framework whose Gaussian Implicit Spatio-Temporal (GIST) blocks replace EM with a learnable GMM, cutting MAE / RMSE by **7.98% / 9.32%** on **2M+** household records.
- Conducted formal analysis benchmarking against **16 baselines** (LSTM, XGBoost, Informer, Autoformer, DLinear, MSGNet) across **8 forecast horizons**—up to **19.4% MAE / 23.7% RMSE** gains at the 96→24-step horizon, confirmed by Diebold–Mariano tests at the **1% level**—and produced the ablation and error visualizations.
  Published: A Coupled Framework for Power Load Forecasting with Gaussian Implicit Spatio-Temporal Block and Attention Mechanisms Network, Computers & Electrical Engineering*, vol. 123, 2025*

Decision Intelligence & Data Analytics Group, Anhui University — Hefei, China
Undergraduate Researcher — *Jun. 2023 – Mar. 2025*

- Developed FSC-DEA, a fairness-aware group decision-making framework combining Probabilistic Linguistic Term Sets, Fehr–Schmidt fairness preferences, and cross-efficiency DEA (Python, NumPy / SciPy).
- Designed an iterative consensus algorithm deriving decision-maker weights from pairwise similarity; raised group consensus from 0.62 to 0.85 in 49 iterations on a case study, with rankings robust to fairness parameters.
  First-author paper under revision: A Fairness-Aware Cross-Efficiency DEA Framework for Multi-Attribute Large-Scale Group Decision-Making

Projects

Proactive User-Profile Inference for Conversational Agents — 2026

- Studied a proactive dialogue agent that asks follow-ups via threshold rules and updates a structured user profile with an LLM reflection module. Reproduced the evaluation (GPT-oss-120B on vLLM; 4 personas × 3 tasks × 5 methods): **profile-recovery accuracy 0.92 vs. 0.38** for a passive baseline; recommendation alignment 0.69→0.86.

Solar Power-Tower Heliostat Field Optimization | *CUMCM (Provincial Award), team of 3* — 2023

- Modeled heliostat-field optical efficiency (cosine, shadowing/blocking via a grid-matrix method, truncation) and optimized field layout with Particle Swarm Optimization (extended to Adaptive PSO); MATLAB.

Skills

Programming Languages: Python, SQL, C++, Java, R, MATLAB, Bash
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, SciPy, NetworkX, Matplotlib, Plotly
Infrastructure & Tools: Git, Linux, Slurm (HPC), Jupyter, MySQL, LaTeX